

From model output to accepted state

A state-preserving architecture for long-context, model-routed, and tool-using AI work.



The work began as an efficiency hypothesis. The experiments rejected the easy story and exposed a more important problem: how probabilistic work crosses into durable truth.

v9 turns that feedback into a control plane: models may explore, compress, and attempt bounded phases, while accepted state advances only after current evidence, exact lineage, verification, and human acceptance.

ACCEPTED' = PROMOTE ? CANDIDATE : ACCEPTED

An unverified attempt can add evidence. It cannot silently replace the last accepted result.

103,788 STRESS CASES / PASS Two byte-identical local passes	256 / 256 FINITE ASSIGNMENTS Production matched the declared oracle	0 ACCEPTED-STATE DRIFT In one 12,012-attempt local run
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The entire project in one page

Core signal. A model output is a proposal. It becomes governed state only through explicit checks, bounded authority, qualified observation, and named acceptance.

What was built

IMPLEMENTED

A linked research stack: Delta Atlas for semantic inspection; a Calibration Ledger and STP for typed transitions; Clutch for bounded model handoffs; BespokeNode for deterministic policy and pre-inference preparation; and a local State Runner for append-only replay.

What the evidence says

TESTED

Early fixed benchmarks showed large input reduction and some quality signals. A later 42-wave repeatability study failed for both Clutch and full history. That failure invalidated a broad efficacy claim and redirected the work toward state preservation, projection sufficiency, and delivery-first measurement.

What is new in practice

TESTED

The system makes acceptance a separate state transition. Candidate work cannot self-promote. Exact content identity, causal order, one terminal decision, current lineage, closed projections, verification, and human acceptance are checked before a result can replace the accepted baseline.

What remains unproved

UNKNOWN

Useful delivery on real projects, independent-team replication, cross-platform qualification, source authentication, browser or provider enforcement, concurrent-writer safety, sudden-power-loss durability, production security, general cost savings, and model-switch protection.

THE SHORTEST OPERATIONAL FORM

```
proposal != permission != execution != effect
effect != observation != acceptance != outcome
```

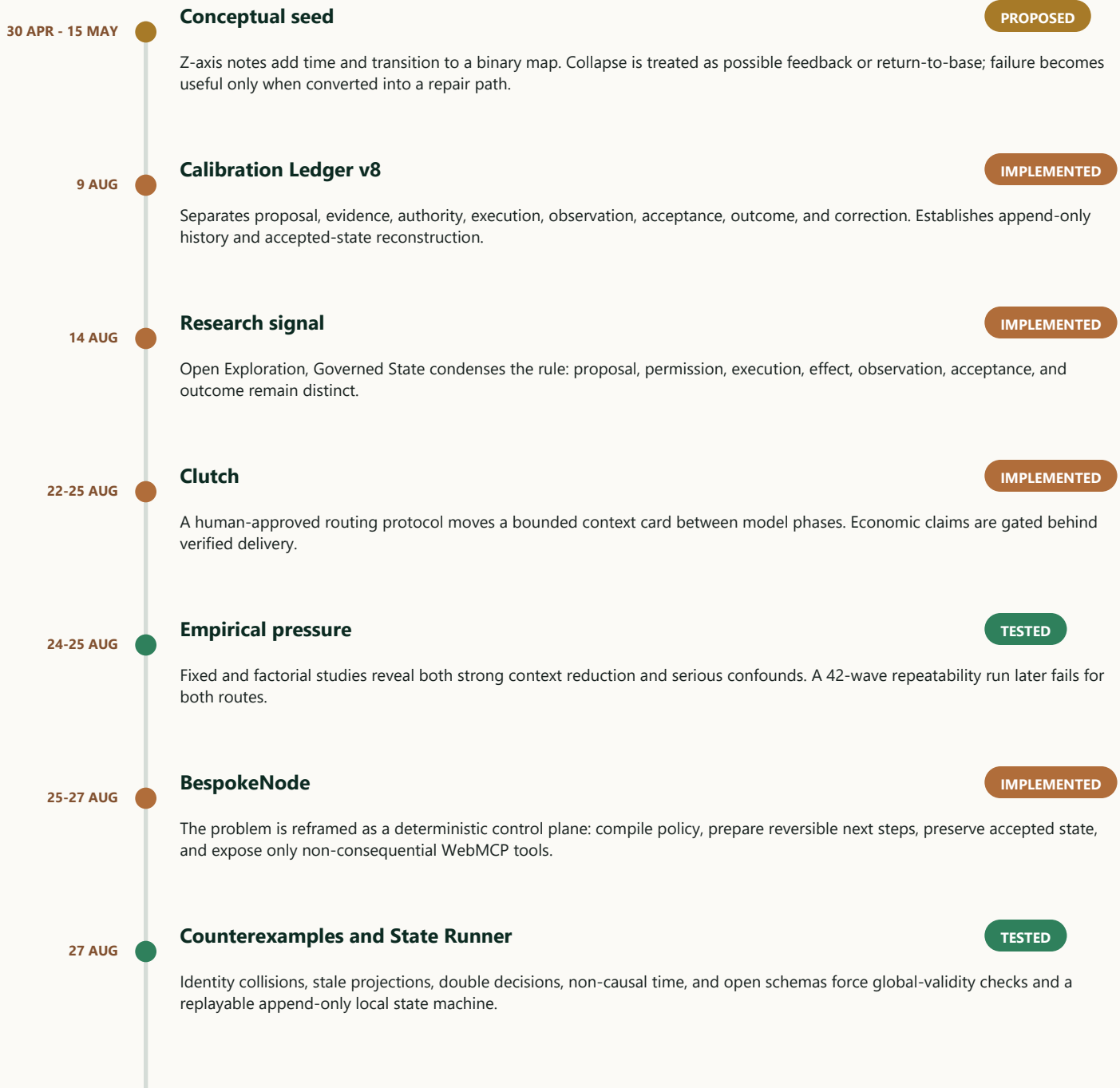
The architecture keeps these records linked without treating them as interchangeable.

CURRENT CLAIM CEILING

A bounded local control plane has been implemented and tested over declared fixtures and one Windows clean-room reproduction. The work supports falsifiable governance and accepted-state preservation inside that model. It does not yet establish product efficacy or production safety.

How the architecture emerged

This is a compressed chronology of the decisions that changed the system. Conceptual notes are labeled as seeds; tests are labeled as evidence.



Chronology: owner research notes; [S1-S2] ledger and orientation; [S3-S11] Clutch and BespokeNode repositories.

The metaphors, translated into testable rules

Metaphors generated questions. They did not count as proof. Their value was retained only when a literal rule, observable failure, or falsifiable test could be written.

Owner shorthand	Literal systems translation	What survived
Z-axis / three trajectories	State includes time, lineage, and future transition constraints, not only a present label.	Causal parent, phase freshness, and accepted-state lineage.
Sovereign zero / return to base	A declared baseline must remain stable while candidates are explored.	Unverified work cannot displace the last accepted state.
Collapse is feedback	A failed attempt can be retained as evidence without being promoted.	Candidate, usage, reason, and fix log remain append-only.
Manual transmission / clutch	Separate reasoning, authorization, execution, and promotion; include a neutral hold.	Exact card approval, finite phases, explicit fuse routes.
Tetris / Operation	Only task-shaped information should pass a strict fit gate; omissions must be detectable.	Projection-sufficiency tests and critical-loss gates.
Oil reaches destination	Cost has no positive standing when the required artifact is not delivered.	Delivery-first economic eligibility.
Metro rails / mirror	Run state and counterexample evidence in parallel, linked by exact identity.	Causal event chain plus quarantine chain and replay.

Discipline. Repeated patterns, model agreement, aesthetic symmetry, and metaphorical fit create candidate questions. Promotion requires literal definitions, observable variables, counterexamples, and a result that another implementation can check.

What was built, layer by layer

01

DELTA ATLAS

IMPLEMENTED

Function-first vocabulary and deterministic inspection of drift, unsupported structure, and projection mismatch.

02

CALIBRATION LEDGER + STP

PROPOSED + TESTED PROFILE

Typed records keep proposal, evidence, authority, action, observation, acceptance, outcome, and correction separate.

03

CLUTCH

IMPLEMENTED

A human-governed protocol for moving bounded phase packages between model profiles while preserving the last accepted artifact.

04

BESPOKENODE

IMPLEMENTED

A local-first policy workbench that compiles user constraints, ranks reversible next steps, and prepares bounded WebMCP drafts.

05

STATE RUNNER

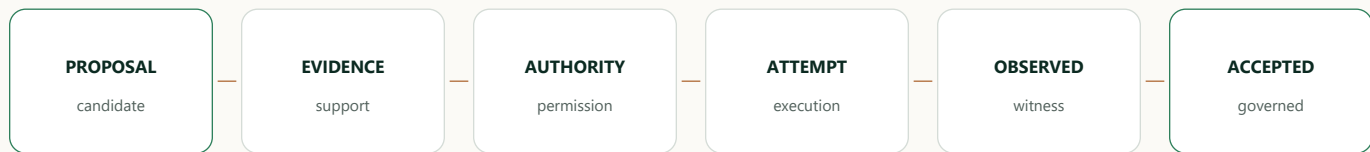
IMPLEMENTED

An append-only local event and quarantine ledger that reconstructs a compact accepted-state projection outside model memory.

The layers do not inherit each other's evidence. A tested reducer does not prove a model is coherent; a routing protocol does not authenticate a tool host; a local state machine does not prove the outside world.

The state-space gap the project addresses

Most agent systems optimize the proposal and tool loop. This project focuses on the seam between a probabilistic candidate and durable organizational state.



STATE VECTOR

```
x_t = (policy, phase, lineage, evidence, effect, recovery,
      verdict, human_decision, accepted_state, telemetry)
```

Each coordinate has a different source and authority. Missing values remain UNKNOWN; they are never rounded into pass or zero.

Local transition state

TESTED

A candidate may be structurally eligible to advance only when the declared finite predicate and global-validity conditions hold. This is the part BespokeNode can test deterministically.

External world state

UNKNOWN

A receipt records what the observation path could see at one boundary. It cannot establish that no material fact existed outside that path, that a witness was honest, or that the later consequence was beneficial.

ADMISSIBILITY, NOT OMNISCIENCE

The architectural question becomes: what reality contact is sufficient for a particular predicate to remain admissible for a particular consequence? The answer is scoped by source, method, freshness, uncertainty, and consequence. A receipt binds that boundary; it does not erase it.

The current baseline algorithm

This is the smallest coherent operational loop shared by the present artifacts. It is deliberately less ambitious than the proposed ten-stage STP lifecycle.

1 INSPECT

Read current policy, accepted-state identity, declared task, constraints, and unknowns. No consequential tool is engaged.

2 COMPILE

Produce one deterministic candidate policy and a small set of reversible pre-inference options.

3 ACCEPT POLICY

A human accepts exact bytes. Acceptance is separate from advice and creates no phase authority.

4 AUTHORIZE PHASE

Bind one finite phase to the accepted policy, current parent state, provider/tool rules, expiry, effects, and recovery class.

5 ATTEMPT

An external worker or adapter performs the bounded work. BespokeNode itself does not execute it.

6 VERIFY + RECORD

Record artifact identity, verifier evidence, verdict, effect, recovery, telemetry, and one terminal decision.

7 PROMOTE OR HOLD

Only a current PASS with global validity and human acceptance advances state. Otherwise preserve evidence and hold.

8 APPEND + REPLAY

Write the event before memory advances; rebuild the projection from the causal chain and checkpoint it.

Invariant 1: accepted state does not drift

The central safety property is small enough to inspect. A candidate can fail, time out, be declined, or remain unknown without silently becoming the baseline.

```
commit_allowed(p) := well_formed(p.delta)
                  AND required_evidence(p) = SUPPORTED
                  AND authority(p) = APPROVED
                  AND policy_version(p) = CURRENT
```

```
NOT commit_allowed(p) => A_after = A_before
```

Ledger v8 rule. This establishes a preservation condition inside the declared transition system; it does not prove that the evidence or policy is wise.

```
ADVANCE = policy_bound AND artifact_evidence
          AND verification_evidence AND effect_recovery_known
          AND verification_passed AND phase_current
          AND lineage_current AND human_accepted
```

BespokeNode finite eligibility surface. All 256 assignments were enumerated twice; production matched the Boolean oracle, and deleting any one factor produced a counterexample.

```
GLOBAL_VALID = exact_content_identity AND canonical_causal_time
              AND one_decision_per_outcome
              AND closed_control_projection
              AND unchanged_projection_epoch
```

```
PROMOTE = GLOBAL_VALID AND ADVANCE
A_next = PROMOTE ? candidate : A
```

Counterexamples showed that eligibility alone was insufficient. Global validity closes identity, history, time, and projection gaps before promotion.

Invariant 2: compression must preserve the decision

A context card is useful only when it retains the information needed for the property under test. Shorter is not automatically better; a projection is too small when it merges histories that require different decisions.

$$e_p = J_p (I - P_b^+ P_b) h_p$$

Interpretation: h_p is source history, P_b is the boundary projection, P_b^+ reconstructs what that projection can represent, and J_p weights task-relevant consequences. A card is sufficient only when the discarded relevant component stays below tolerance and no critical component is lost.

Exact identity

TESTED

Bind the exact policy, phase, parent state, card, evidence, and result. Short display IDs are correlation aids only.

Decision parity

TESTED

Group histories that share a projection. If any group contains both PROMOTE and HOLD, the projection is insufficient.

Task sufficiency

PROPOSED

Run held-out work with matched acceptance criteria. This remains the empirical layer the finite matrix cannot replace.

COUNTEREXAMPLE-GUIDED MAINTENANCE

Counterexample	Unsafe merge	Smallest retained repair
32-bit receipt collision	Different accepted states shared one display identity.	Full SHA-256 transition identities.
PASS plus decline	Independent log parsing allowed two terminal decisions.	Aggregate one-decision validation.
Backdated child	Host-dependent time could precede its accepted parent.	Canonical UTC and parent chronology.
Open WebMCP state	Unknown fields could widen the control projection.	Closed allowlisted schema.
Stale phase draft	Policy changed between read and draft application.	Projection epoch and causal reread.

Source: [S9] Finite Transition and Projection-Sufficiency Research.

Invariant 3: count efficiency only after delivery

The oil-tanker rule became an explicit constraint: token, cache, time, and dollar measurements can describe a failed route, but they cannot be promoted as successful system efficiency unless the required cargo arrives safely.

```
SAFE_DELIVERY = task_complete AND required_checks_pass
                AND no_policy_violation
                AND accepted_state_preserved
```

```
EFFICIENCY_ELIGIBLE = SAFE_DELIVERY AND telemetry_complete
```

Missing telemetry is UNKNOWN, never zero. Failed attempts remain charged. Human review time is reported separately until it can be measured credibly.

AMORTIZATION, STATED HONESTLY

```
D_b(n) = planning_and_card_cost
        + SUM_p<=n (clutch_worker_cost_p - history_worker_cost_p)
```

break-even at boundary b when $D_b(n) \leq 0$

A front-loaded advisor investment can amortize over several bounded phases only if delivered worker phases are cheaper than full-history phases by enough to repay card creation and review.

OPTIMIZATION IS CONSTRAINED

```
minimize total_route_cost
subject to delivery_quality >= frozen_floor
           critical_losses = 0
           final_delivery = PASS
```

Penalty scores may diagnose a route, but hard delivery and critical-loss gates control promotion.

The empirical program, without inheritance

Each study answers a narrower question. Later failures reduce the claim ceiling; they are not averaged away by earlier positive signals.

Study	Unit / design	Observed signal	Claim ceiling
Clutch v1 local pilot	Six rounds; private minimized packet	6 accepted Clutch artifacts vs 5 direct; fewer raw tokens and prompt bytes; longer elapsed time.	Descriptive only. No efficiency winner because the direct arm failed its quality bar.
Fixed v2 confirmatory	15 synthetic tasks; 3 templates; 1 advisor; 241 provider requests	95.93%-95.94% allocated input reduction; average quality met -10 point noninferiority; delivery mismatch remained.	Fixed set only. Model-switch resilience unresolved.
Factorial coherence	192 frozen cells; full history vs compiled card; Terra/Luna phases	Fewer collapses and lower long-context input with cards; primary switch contrast undefined.	Private exploratory feasibility. INCONCLUSIVE.
42-wave repeatability	42 primary synthetic waves plus one alternate advisor	2 deliveries for Clutch and 2 for full history; \$43.43 total accounted spend.	REPEATED_FAIL_DELIVERY. No positive reliability or savings claim.
BespokeNode local gate	Finite predicates, hostile fixtures, stress matrices, State Runner	Deterministic local conformance and accepted-state preservation in declared cases.	Control-plane behavior only. No product efficacy inheritance.

Claim ceiling - no inheritance. A positive input-reduction result does not prove delivery. A deterministic control-plane pass does not rescue a failed routing study. BespokeNode is evaluated for state transition integrity, not credited with Clutch efficacy.

Fixed benchmark: strong reduction, mixed delivery

TL;DR - what we actually did. We gave the same synthetic multi-phase builds either full history or an approved compact card, kept the worker route stable or switched it from Terra to Luna, and measured hidden-probe quality, strict completion, transmitted input, and provider charges.

241

PROVIDER REQUESTS

1 advisor + 240 workers

\$16.08

PROVIDER CHARGE

human review unmeasured

95.93%

STABLE INPUT REDUCTION

allocated advisor included

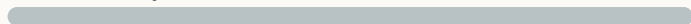
95.94%

SWITCHED INPUT REDUCTION

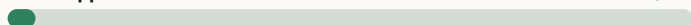
allocated advisor included

TRANSMITTED INPUT

Stable full history 5,049,824

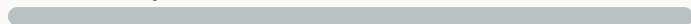


Stable approved cards + advisor 205,301



STRICT FINAL COMPLETION

Stable full history 15 / 15



Stable cards 9 / 15



Switched full history 10 / 15



Switched cards 13 / 15



What passed

TESTED

On average, the approved-card treatment met the preregistered -10 percentage-point noninferiority margin on the fixed hidden-probe score. No declared boundary violation or collapse was recorded. Provider input and measured charges were substantially lower in this fixed set.

What did not follow

UNKNOWN

Average-score noninferiority did not preserve strict completion in the stable route. The tasks were fifteen deterministic variants of three templates, with one advisor realization. Switch resilience was classified NEUTRAL_OR_UNRESOLVED. Human review time was not measured.

The game-changing feedback was methodological: an aggregate quality pass and a 95.9% input reduction can coexist with a delivery regression. Delivery therefore became a separate gate, not one more score inside an average.

Factorial pilot: cards helped under pressure, primary test inconclusive

TL;DR - what we actually did. We varied context length, card versus full-history carriage, task family, phase, and worker model across 192 frozen cells to ask whether already-correct behavior survived a handoff and whether long context itself caused operational collapse.

Primary result

UNKNOWN

INCONCLUSIVE. No previously passing probe regressed at either switch boundary, so the aggregate switch penalty was zero and most task-level retention contrasts were undefined. Zero observed regression was not promoted into proof of harmless switching.

Operational result

TESTED

Compiled cards had fewer collapsed cells in every fixed task family and a higher descriptive active-probe pass rate. This combined compression with clearer structure, so the causal ingredient was not isolated.

ALL 192 FROZEN CELLS

Full-history collapses 35 / 96



Card collapses 12 / 96



History active probes 234 / 384 | 60.9%

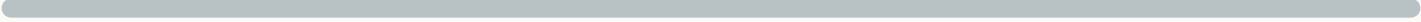


Card active probes 271 / 384 | 70.6%



LONG-CONTEXT PHASE 3 - LUNA WORKER

Full-history average input 81,452 tokens



Compiled-card average input 1,293 tokens

**83.5%**

GENERATION-CHARGE REDUCTION

Long-context Phase 3 slice only

24

INCOMPLETE RESPONSES

All hit the 6,000-token ceiling

23

WORKER HOLD RESPONSES

21 occurred in Phase 1

Operational completion and artifact correctness had to remain separate: one incomplete provider response still left source that passed every active probe. That observation later shaped the outcome ledger and fix-log design.

The repeatability failure that changed the project

TL;DR - what we actually did. We repeated fresh near-, middle-, and deep-context waves to test whether successful end-to-end delivery repeated, not merely whether average scores looked good. Frozen decision: REPEATED_FAIL_DELIVERY. Clutch delivered twice and full history delivered twice.



\$43.43

TOTAL PROVIDER-ACCOUNTED SPEND

Inside a \$64 ceiling; no positive economic conclusion

1

RECORDED RECOVERY EVENT

Retained as feedback, not promoted as success

POST-HOC SEMANTIC AUDIT

Two exploratory reducers found representation-sensitive matches that the strict scorer missed: 27/42 for Clutch and 34/42 for full history. The observed difference was -0.166667, and neither proposed quality gate cleared. Because aliases were chosen after the run, the frozen strict result remains authoritative.

WHAT THE FAILURE ACTUALLY BOUGHT

Old assumption	Observed defect	Architectural response
Good average quality is enough	Strict cargo often did not arrive.	Make delivery a co-primary hard gate.
Failure data is inherently valuable	Raw failures repeated without a mechanism.	Require a fix log, named repair, and later test.
One score defines outcome	Representation and completion disagreed.	Keep transport, artifact, verifier, and acceptance states separate.
More attempts create confidence	Repeated low delivery only increased spend.	Use preregistered futility stops and no hidden retries.
A card is compression	Structure and selection were confounded.	Ablate full history, automatic card, reviewed card, and worker model.

The failure did not become a win. The fix logs became the bridge from a failed efficacy hypothesis to a testable state-transition architecture.

Sources: [S6] frozen repeatability analysis; [S7] post-hoc semantic audit and unrun mirror-study proposal.

Feedback became architecture only after repair

The project treats a defect as useful only when it produces a minimal repair, a retained counterexample, and a rerun of the frozen gate.

Feedback / defect	Repair	Retest
Zero-call tool leakage	Closed tool authority and explicit no-tool states.	Hostile direct tests.
Rounded budget comparison	Compare exact values; display rounding cannot govern.	Policy and routing stress matrix.
Profile or card substitution	Canonical reconstruction and exact SHA-256 identities.	Altered-card and cross-policy replay rejection.
32-bit receipt collision	Full SHA-256 on transition receipts.	Frozen collision pair retained as counterexample.
PASS and decline on same outcome	Aggregate snapshot parser enforces one terminal decision.	Acceptance replay and disposition tests.
Stale or backdated phase	Canonical UTC, parent chronology, phase freshness, lineage checks.	Temporal and sibling-race probes.
Open WebMCP projection	Strict schema and separately authored contract oracle.	10 ordinary + 10 hostile fixtures.
State stored in model context	Append-only local event/quarantine chains plus deterministic projection.	12,012-attempt replay and recovery run.
Clean-room LF/CRLF drift	Repository-level LF normalization and exact blob comparison.	Candidate 002+ Windows attestations.

counterexample -> preserve witness -> add smallest separating check
-> rerun frozen matrix -> narrow claim ceiling

This is maintenance, not proof by accumulation. A new counterexample can still require a new factor.

Sources: [S8-S11] BespokeNode README, red-team report, finite-transition research, and attestations.

BespokeNode Candidate 004: what now exists

TL;DR - what we actually tested. We attacked the local control plane with ordinary, hostile, combinatorial, identity, timing, lineage, and replay cases to see whether any invalid candidate could silently become accepted state.

86

FOCUSED TESTS

Complete local gate

20

WEBMCP FIXTURES

10 ordinary + 10 hostile

103,788

STRESS SCENARIOS / PASS

Two byte-identical passes

256

FINITE ASSIGNMENTS / PASS

Eight drop-one witnesses

Deterministic policy workbench

IMPLEMENTED

Compiles privacy, provider, tool, network, evidence, budget, and interaction constraints into one inspectable candidate. Route Ahead ranks exactly three reversible advisory options. Working modes alter presentation only; they cannot widen authority or privacy.

Bounded acceptance chain

TESTED

Policy acceptance, phase authorization, outcome recording, and result promotion are separate receipts. A phase is consumed by PASS, FAIL, or UNKNOWN. Failed, unknown, stale, altered, or declined outcomes remain evidence and cannot advance the linear accepted-state chain.

CANDIDATE IDENTITY

```
baseline: BESPOKE_NODE_LOCAL_0_3_0_CANDIDATE_004
commit: 4c03cd7a5b8badc7f15d38a5d1f33e7713b6e11c
tree: 583943ea99ec48edd4def92f0a32821a171597ef
manifest: a05e8082848d4f30...3e3718fa6a7f1750
```

One owner-approved Windows clean-room attestation reproduced exact source bytes and the full local gate. The successful run reused an identified local dependency snapshot after the offline cache lacked ws@8.21.0.

Claim ceiling. Local conformance and state preservation in declared fixtures. No useful-outcome, production-security, cross-platform, dependency-provenance, browser-host, or independent-team claim.

Running state outside model memory

TL;DR - what we actually tested. We fed thousands of valid and duplicate observations through an append-only local runner, closed and replayed it, damaged its tail, and separately sent normal and hostile calls through the bounded WebMCP surface.

S(n) = reduce(S(n-1), E(n))

Sequence and previous-event SHA-256 define causal order. Timestamps are metadata and cannot create authority.

write ledger -> then advance memory

If append fails, in-memory state remains at the prior projection. Damaged replay falls back to a self-valid checkpoint in read-only recovery.

12,012

ATTEMPTS

12,000 valid + 12 duplicate

549.82/s

OBSERVED THROUGHPUT

One Windows attestation run

0.5349 ms

P95 APPEND LATENCY

Local observation only

0

SILENT LOSS / DRIFT

Exact reopen and replay parity

TWO LINKED RAILS

Valid event chain TESTED

Closed observations with current stream, unique source identity, and valid causal parent update the deterministic projection. Accepted-state observations additionally require PASS, an allowlisted source, and the current accepted parent.

Quarantine chain TESTED

Malformed, duplicate, wrong-stream, stale-parent, or unauthorized attempts are recorded by digest and reason on a separate append-only chain. Quarantine changes the operational view but never the accepted state.

WEBMCP SURFACE: PREPARATION WITHOUT PROMOTION

Tool	Permitted function	Explicitly absent
inspect_control_state	Return a minimized, closed projection.	Acceptance, execution, mutation.
prepare_policy_draft	Prepare a visible candidate policy.	Policy acceptance or hidden authority.
prepare_phase_request	Prepare a finite request bound to current state.	Worker dispatch or state promotion.

The present surface gives agents safe ways to inspect and prepare. Consequential authority remains outside the agent tool interface. Actual host retraction, identity, and enforcement still require browser-host qualification.

Sources: [S8] BespokeNode README; [S10] State Runner; [S11] attestation; [S12] WebMCP draft.

What is established, and where the edge is

L0	Declared design Schemas, invariants, threat model, stop conditions.	IMPLEMENTED
L1	Local conformance Deterministic fixtures, counterexamples, exact replay, release boundary.	TESTED
L2	One-host reproduction Exact Candidate 004 source bytes and local gate on one Windows host.	TESTED
L3	Two-host / independent implementation Fresh macOS/Linux, real browser hosts, independently authored verifier.	UNKNOWN
L4	Useful comparative outcome Held-out real tasks, matched controls, delivery, human effort, total cost.	UNKNOWN
L5	Production assurance Authenticated adapters, transactional state, concurrency, power loss, security review.	UNKNOWN

The edge is mapped by the first unresolved predicate, not by confidence or enthusiasm. A local pass remains valuable at L1-L2 and must not be narrated as L4-L5.

The next falsifiable program

The architecture is ready for external scrutiny, but internal testing is not finished. The next work should maximize information gained per unit of money, reviewer time, and implementation effort.

Checkpoint	Design	Pass condition	What a failure means
1. External replay	Independent reviewer runs the exact local gate on fresh bytes.	Same manifest, fixtures, and receipts; no source edits.	Packaging or hidden-environment dependence.
2. Host conformance	Run three bounded WebMCP tools in ChatGPT and WebMCP-enabled Chrome.	Exact tool discovery, closed schemas, no acceptance or mutation path.	Adapter/host semantics violate the local model.
3. Lifecycle stress	Single writer with forced termination at every write boundary.	No silent promotion, replay mismatch, or state loss beyond declared recovery.	Persistence ordering is insufficient.
4. Projection experiment	Matched held-out tasks: full history, automatic card, reviewed card; same active spec.	Predeclared noninferiority, absolute delivery floor, zero critical losses.	Projection is too small or treatment confounded.
5. Comparative economics	Count all calls, failures, human review, repair, latency, and infrastructure.	Only evaluate after both arms clear delivery and telemetry gates.	No defensible efficiency claim.
6. Authenticated adapter	Disposable real tool host with signed receipts and compare-and-set promotion.	Reject stale, forged, concurrent, and unauthorized transitions.	Control plane is advisory, not enforced.

STOP RULES

- Stop if the next step would publish, deploy, spend money, expose private material, or claim novelty without fresh authority.
- Stop when a result has no next falsifiable checkpoint; do not replace evidence with more narrative.
- Keep the mathematical flagship lane separate from application evidence. Metaphor and model agreement do not promote a theorem candidate.

Near-term objective. Establish that two independent hosts preserve the same bounded transition semantics. Then test whether the compact projection improves real delivery under matched task requirements. Security and economic claims wait behind those gates.

What each method actually does

These components overlap, but they solve different problems. Calling all of them state machines would erase their practical value.

Method	Primary job	Other functions it performs	Current ceiling
Pin Check	Audit whether a claim or evaluator label is actually supported by a pinned source and rule.	Detects label absorption, threshold gaming, evaluator drift, unexplained rounding, and convenient reclassification. Its numeric form returns TRACED, ROUNDED, or UNDEFINED instead of guessing.	Method demonstration plus bounded experiment note. Not yet a packaged, independently replicated skill.
Delta Atlas	Translate terms by function and expose semantic gaps.	Crosswalks meanings, identifies drift and unsupported equivalence, and gives humans a deterministic inspection surface before model judgment.	Implemented public case study; not a universal semantic validator.
Calibration Ledger / STP	Keep proposal, evidence, authority, action, observation, acceptance, outcome, and correction distinct.	Provides provenance, append-only corrections, explicit unknowns, replay rules, and a shared vocabulary for handoffs and audits.	Ledger implemented in bounded profiles; full ten-stage lifecycle remains proposed.
Clutch	Move bounded work across model or context boundaries without letting the handoff self-authorize.	Selects context, separates expensive judgment from routine work, freezes phase contracts, governs human approval, limits retries, names recovery routes, preserves fix logs, supports cross-harness packaging, and measures quality before economics.	Protocol and evaluation kit implemented. General delivery, savings, and model-switch protection remain unproved.
BespokeNode	Compile a bespoke local policy before inference or tool use.	Combines human interaction fit, privacy, provider, budget, tool, and evidence constraints; ranks reversible next moves; prepares bounded WebMCP drafts; separates policy, phase, outcome, and promotion.	Local Candidate 004 conformance only.
State Runner	Keep durable operational state outside any model's context window.	Sequences events, quarantines invalid attempts, rebuilds a compact projection, checkpoints, replays, and enters read-only recovery rather than inventing state.	Single-writer local experiment; no authentication, concurrency, or power-loss assurance.
Open Mirror	Generate candidate questions through multiple literalized lenses.	Helps escape tunnel vision, records where an analogy breaks, and returns useful observations to the governed baseline without treating recurrence as evidence.	Exploration only; never a promotion mechanism.

Pin Check lesson. In the source demonstration, 118 labeled decisions showed that easy defensible labels absorbed 75%-100% of decisions while the strongest labels received 0%-2%. Tightening one label often moved traffic into another. The bounded lesson is to compute deterministic comparisons when possible and leave unsupported cases UNDEFINED. This is not independent replication of the original PDF.

Every major test in plain English

Each row answers two questions: what were we actually doing, and what did that test earn the right to say?

Test	TL;DR - what we did	What the result means
Pin Check demonstration	Changed labeling pressure across repeated decisions to see whether the evaluator used the strongest justified label or flowed toward easier labels.	Supports label-absorption as a testable concern; strongest labels rarely fired in the bounded source study. Needs executable packaging and replication.
Clutch v1 local pilot	Ran the same six-round build through a governed card route and a growing-context route, then checked delivered artifacts and measured prompt volume and elapsed time.	A useful early signal, but the direct arm missed its quality bar, so the study correctly refused to name an efficiency winner.
Fixed v2 confirmatory	Compared full history with human-approved cards on 15 synthetic multi-phase tasks, with stable and switched workers.	Large input reduction and average-score noninferiority on this fixed set; strict delivery and switch protection were not established.
Factorial coherence	Crossed context length, carriage method, task family, phase, and worker model across 192 cells.	Cards collapsed less under long-context pressure; the preregistered switch-retention question remained inconclusive.
42-wave repeatability	Repeated fresh near/middle/deep context waves and counted verified end-to-end deliveries.	Both routes delivered 2/42. This falsified the broad repeatability story and forced delivery-first architecture.
Semantic audit	Re-scored the frozen repeatability outputs with two exploratory meaning-aware reducers to locate representation-sensitive failures.	Found format sensitivity, but post-hoc aliases did not rescue the frozen result or clear proposed gates.
Transition projection	Flipped all eight promotion factors through every 0/1 combination, compared code to an oracle, then removed one factor at a time.	256/256 local parity; every factor was necessary inside the declared model. This is bounded sufficiency, not universal minimality.
WebMCP hostile suite	Sent 10 ordinary and 10 hostile preparation/inspection calls through production code and a separately authored contract oracle.	Exact local agreement and accepted-state preservation. Real host enforcement remains unqualified.
State Runner stress	Processed 12,000 valid events plus duplicate attempts, persisted, reopened, replayed, and checked the accepted projection.	No silent loss, duplicate promotion, replay mismatch, or accepted-state drift in one local run.
Windows clean room	Cloned exact source bytes, compared them with commit blobs, restored the dependency graph, and ran the full gate.	One-host source and gate reproduction. Dependency independence was not established because one package was absent from the offline cache.

A test result earns only its named sentence. Positive signals do not inherit through failed gates, and later counterexamples narrow earlier claims.

Test sources: [S3-S11] plus [S17].

Sources, identities, and how to review this ledger

This document is a shareable orientation and evidence map. It is not a substitute for the source artifacts, raw results, frozen manifests, or an independent reproduction.

ID	Artifact	Locator / role
S1	Calibration Ledger v8	Local canonical PDF: work/stp_v1_2_candidate/source_render/ledger/Calibration_Ledger_v8_source.pdf
S2	Open Exploration, Governed State	output/research-signal/RESEARCH-SIGNAL-GUIDE.md
S3	Clutch v0.4 research preview	https://github.com/JakeTOpenSource/clutch-skill
S4	Fixed v2 confirmatory summary	work/clutch-v2-benchmark-pr/evidence/clutch-v2-confirmatory-001/research-summary.json
S5	Factorial coherence handoff	output/clutch-factorial-coherence-v1-handoff.md
S6	42-wave repeatability analysis	output/clutch-v2-repeatability-001/run/analysis.json
S7	Semantic audit + mirror proposal	output/clutch-v2-repeatability-semantic-audit-001/
S8	BespokeNode Candidate 004	work/build-a-moat/README.md; commit 4c03cd7a5b8badc7f15d38a5d1f33e7713b6e11c
S9	Finite transition research	work/build-a-moat/docs/FINITE-TRANSITION-RESEARCH.md
S10	State Runner	work/build-a-moat/docs/STATE-RUNNER.md
S11	Candidate 004 attestation	work/bespoke-node-attestations/cleanroom-windows-4c03cd7-pass.json
S12	WebMCP draft	https://webmachinelearning.github.io/webmcp/
S13	JSON Canonicalization Scheme	RFC 8785 - https://www.rfc-editor.org/rfc/rfc8785
S14	Protection principles	Saltzer and Schroeder - https://web.mit.edu/saltzer/www/publications/protection/Basic.html
S15	Safety and liveness	Lamport - https://www.microsoft.com/en-us/research/publication/proving-possibility-properties/
S16	AI risk-management context	NIST AI RMF 1.0 - https://doi.org/10.6028/NIST.AI.100-1
S17	Pin Check method note	clutch-qc-eval-v20/sources/pin-check-method-note.md; source PDF SHA-256 37e7da7118f972e4bfbae71a0573c1abf70d02ac38b074d486594b2e85924f4b

The source register is intentionally exact enough to locate the evidence and narrow enough to avoid private raw prompts, hidden probes, and secrets. Review and contribution guidance follows on the final page.

How to review and reproduce the work

The fastest serious review begins with exact bytes and local behavior, then moves outward to empirical delivery and host enforcement.

Review order

1. Reproduce the Candidate 004 local gate. 2. Inspect the finite predicate and counterexamples. 3. Read the Clutch studies in chronological order. 4. Treat every claim above its cited ceiling as unestablished.

Contribution boundary

Jake Tiller initiated the program, supplied the conceptual framing and acceptance boundaries, directed experiments, and owns release decisions. AI systems assisted with research, drafting, coding, testing, and adversarial review; their agreement is not independent validation.

MINIMAL INDEPENDENT REVIEW PACKET

- Exact Candidate 004 commit, tree, manifest, source-set digest, and one-host attestation.
- The 256-case promotion matrix, eight drop-one witnesses, and five global-validity counterexamples.
- The frozen Clutch repeatability analysis, including failed deliveries and all charged attempts.
- A reviewer statement that separates reproduced local behavior from product efficacy and production assurance.

Contact goal. The strongest useful response is not agreement. It is one exact reproduction, one counterexample the present gate misses, one better-matched empirical design, or one clearly bounded integration test on a different host.

Document status: Calibration Ledger v9, shareable research summary, prepared 27 August 2026. Not peer reviewed. Document text licensed CC BY 4.0; software remains under each repository's license. Private raw prompts, hidden probes, secrets, and unpublished task content are excluded.

One-line handoff: Keep exploration open. Keep promotion explicit. Preserve the accepted state, the counterexample, the unknowns, the authority boundary, and the people affected by the result.